

STATSIG lecture 2

Analysis of random signals in time domain: correlation functions

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

Random signals and random variables

Random signal $X(t)$ – contains random/stochastic components with some probability distribution

Signal values at a fixed time instant t_i , $X(t_i)$ is a random variable across all possible outcomes of the signal $X(t)$ – the "*ensemble*"

Autocorrelation function for random signal X(t)

General definition $R_{xx}(t_1, t_2) = E[X(t_1)X(t_2)]$

Measures the correlation between zero-mean signal value at two times

If X(t) is not zero-mean, the autocovariance function is used instead

Often X(t) is assumed to be WSS = Weak Sense Stationary
(constant mean and correlation function only depending on "lag")

WSS definition $R_{xx}(\tau) = E[X(t)X(t + \tau)]$ where lag $\tau = t_2 - t_1$

The operator correlation is closely related to the operator convolution

Convolution is "Flip and Slide" (here convolution of signal with itself)

Correlation is "No Flip but Slide" (here correlation of signal with itself)

Follows: MATLAB examples and general properties

Ensemble average (theoretical) vs. Time average (practical)

Ensemble mean (average)

$$\mu_x = E[X(t)]$$

Can only be calculated if probability density function for the whole ensemble is known (usually not the case for practical signal)

Time average (estimated) mean

$$\hat{\mu}_x = A[X(t)] = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X(t) dt$$

Requires only one measured signal (should be "representative" for all possible signals)

Ensemble autocorrelation function

$$R_{xx}(\tau) = E[X(t)X(t + \tau)] \quad \text{where } \underline{\text{lag}} \quad \tau = t_2 - t_1$$

Can only be calculated if multivariate probability density function for the whole ensemble is known (usually not the case for practical signal)

Time average (estimated) autocorrelation function

$$\hat{R}_{xx}(\tau) = A[X(t)X(t + \tau)] = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X(t)X(t + \tau) dt$$

Requires only one measured signal (should be "representative" for all possible signals)

MATLAB (discrete-time) calculation of autocorrelation function

Discrete-time estimation of autocorrelation function for N signal samples

$$\hat{R}_{xx}(\tau) = c \sum_{n=0}^{N-1} X(n)X(n + \tau) \quad \text{for } \tau \geq 0 \quad \text{and with} \quad \hat{R}_{xx}(-\tau) = \hat{R}_{xx}(\tau)$$

MATLAB syntax: $[R_{xx}, \text{lags}] = \text{xcorr}(X, \text{'option'})$

The 'option' parameter decides sum normalization

'option' = 'none':	c = 1
'option' = 'coeff':	R _{xx} (0) = 1
'option' = 'biased':	c = 1/N
'option' = 'unbiased':	c = 1/(N-lag)

Follows: examples

Crosscorrelation function for random signals $X(t)$, $Y(t)$

Two correlated signals $X(t)$, $Y(t)$

For example in a radar system $X(t)$ could be the sent signal while $Y(t)$ is the delayed, attenuated and noisy received (reflected) signal

WSS definition $R_{xy}(\tau) = E[X(t)Y(t + \tau)]$ where again τ is the lag

Measures the correlation between samples of each of the two signals as a function of lag = time-difference between samples

Time average (estimated) crosscorrelation function

$$\hat{R}_{xy}(\tau) = A[X(t)Y(t + \tau)] = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X(t)Y(t + \tau) dt$$

MATLAB syntax: $[R_{xy}, lags] = xcorr(X, Y, 'option')$

Follows: MATLAB examples and general properties